

1. Create a **CAR** class, which is **abstract** super class of all cars.
 - a) CAR class includes TWO attributes String **Company** and **serialNumber**.
 - b) A **constructor** to initialize all the data members of **CAR** class.
 - c) Override **toString()** method to return details about the **CAR** class.

2. Create an **interface CarProperty** with the following details:
 - a) Declare a constant **SmartParking** of type **Boolean** with value **true**.
 - b) Declare a method called **ParkingSupport()** which returns a **String** value.

3. Create the **SmartCar** class that inherits **CAR** class and implements **CarProperty** interface:
 - a) **SmartCar** class must include TWO attributes to store the **modelYear** of type **integer** and **count** of type **static int**.
 - b) Define a **constructor** that takes appropriate parameters so as to initialize data members of **CAR** class as well as **SmartCar** class.
 - c) When you override **ParkingSupport()** method, it will return "**This car support smart Parking**" if the **modelYear** over than **2018** otherwise it should return "**This car doesn't support smart Parking**".
 - d) Override **toString()** method to return details about the **SmartCar** class, which also includes the details of **super class**.

4. Create **CarTester** class to test your application:
 - a) Declare and instantiate a variable **myCar** to be an empty **ArrayList** of type **SmartCar**. [All the data must be entered by **the user**. The **size** of **ArrayList** depends on the first number of **the user entering** it. Use **Exception** handling technique to make sure your code doesn't terminate abnormally. Use do-while loop to let user enter correct **model_Year** data in case wrong data is entered.]
 - b) List all **myCar** object details by invoking appropriate methods.
 - c) Using the concept of **File I/O**, Write your complete output car details in a **file** named "**car.txt**".

Sample put:

```
Enter the number of Cars for the day :
2 [ENTER]
Enter the name of Company number[ 1 ]
MG [ENTER]
Enter the serialNumber :
123abc [ENTER]
Enter the Model :
S [ENTER]
!!! The serialNumber must be number only !!!
Try again
Enter the Model :
2022 [ENTER]
Enter the name of Company number[ 2 ]
BMW [ENTER]
Enter the serialNumber :
98765 [ENTER]
Enter the Model :
2018e [ENTER]
!!! The serialNumber must be number only !!!
Try again
Enter the Model :
2018 [ENTER]
```

BUILD SUCCESSFUL (total times: 1 minute 22 seconds)

The file output:

```
Car.txt - Notepad
File Edit Format View Help
Final lab exam :
-----
Company      SerialNumber  Model_Year
MG           123abc       2022      This car support smart Parking
BMW          98765       2018      This car doesn't support smart Parking
-----
Total number of cars : 2
```